

Vehicle Service Management System

Code

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import mysql.connector
from datetime import datetime
from reportlab.platypus import SimpleDocTemplate, Paragraph, Spacer
from reportlab.lib.styles import getSampleStyleSheet
import os

# ----- DATABASE CONNECTION -----
try:
    conn = mysql.connector.connect(
        host="127.0.0.1",
        user="root",
        password="abha123",
        database="vehicle_service_db"
    )
    cursor = conn.cursor()
    print(" Connected to MySQL")
except Exception as e:
    print("Database Error:", e)
    exit()

# ----- CUSTOMER MODULE -----
def add_customer():
    name = input("Name: ")
    phone = input("Phone: ")
    vehicle = input("Vehicle Number: ")
    address = input("Address: ")
    purchase_date = input("Purchase Date (YYYY-MM-DD): ")
    km = int(input("Current KM: "))
    cursor.execute("""INSERT INTO customers
(name, phone, vehicle_number, address, purchase_date, current_km)
VALUES (%s,%s,%s,%s,%s,%s)""",
(name, phone, vehicle, address, purchase_date, km))
    conn.commit()
    print("Customer added! ID:", cursor.lastrowid)

def view_customers():
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        cursor.execute("SELECT * FROM customers")
        for row in cursor.fetchall():
            print(f"ID:{row[0]} | Name:{row[1]} | Vehicle:{row[3]} | KM:{row[6]}")

def search_customer():
    val = input("Enter Phone or Vehicle: ")
    cursor.execute("SELECT * FROM customers WHERE phone=%s OR vehicle_number=%s", (val, val))
    for row in cursor.fetchall():
        print(row)

def update_km():
    vehicle = input("Vehicle Number: ")
    km = int(input("New KM: "))
    cursor.execute("UPDATE customers SET current_km=%s WHERE vehicle_number=%s", (km, vehicle))
    conn.commit()
    if cursor.rowcount > 0:
        print("KM Updated!")
    else:
        print("Vehicle not found!")

# ----- SERVICE MODULE -----
def book_service():
    vehicle = input("Vehicle Number: ")
    cursor.execute("SELECT customer_id FROM customers WHERE vehicle_number=%s", (vehicle,))
    data = cursor.fetchone()
    if not data:
        print("❌ Vehicle not found!")
        return
    cid = data[0]
    stype = input("Service Type: ")
    mech = input("Mechanic: ")
    date = input("Date (YYYY-MM-DD): ")
    cursor.execute("""INSERT INTO services
(customer_id, service_type, mechanic, service_date, status)
VALUES (%s,%s,%s,%s,'Pending')""",
(cid, stype, mech, date))
    conn.commit()
    print("Service Booked!")

def update_service():
    sid = input("Service ID: ")
    status = input("Status (Pending/In Progress/Completed): ")
    cursor.execute("UPDATE services SET status=%s WHERE service_id=%s", (status, sid))
    conn.commit()

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        if cursor.rowcount > 0:
            print("Updated!")
        else:
            print("Service not found!")

def service_history():
    vehicle = input("Vehicle Number: ")
    cursor.execute("""
SELECT c.name, s.service_type, s.mechanic, s.service_date, s.status
FROM customers c
JOIN services s ON c.customer_id = s.customer_id
WHERE c.vehicle_number=%s
""", (vehicle,))
    data = cursor.fetchall()
    if not data:
        print("No history found.")
    else:
        for row in data:
            print(f"Name:{row[0]} | Service:{row[1]} | Mechanic:{row[2]} |
Date:{row[3]} | Status:{row[4]}")

# ----- SPARE PARTS -----
def add_part():
    name = input("Part Name: ")
    qty = int(input("Quantity: "))
    price = float(input("Price: "))
    cursor.execute("INSERT INTO spare_parts VALUES (NULL,%s,%s,%s)", (name, qty,
price))
    conn.commit()
    print("Part Added!")

def view_parts():
    cursor.execute("SELECT * FROM spare_parts")
    for row in cursor.fetchall():
        print(f"ID:{row[0]} | {row[1]} | Qty:{row[2]} | ₹{row[3]}")

def update_part_stock():
    try:
        part_id = int(input("Enter Part ID: "))
        cursor.execute("SELECT part_name, quantity FROM spare_parts WHERE part_id=%s",
(part_id,))
        part = cursor.fetchone()
        if not part:
            print("Part not found!")
            return
        name, current_qty = part
        print(f"Current Stock of {name}: {current_qty}")

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        change = int(input("Enter quantity to ADD (+) or REMOVE (-): "))
        new_qty = current_qty + change
        if new_qty < 0:
            print("Stock cannot be negative!")
            return
        cursor.execute("UPDATE spare_parts SET quantity=%s WHERE part_id=%s", (new_qty,
part_id))
        conn.commit()
        print(f"✅ Stock updated! New Quantity: {new_qty}")

        # LOW STOCK ALERT
        if new_qty < 5:
            print(f"LOW STOCK ALERT: {name} only {new_qty} left!")
        if new_qty == 0:
            print(f"{name} is OUT OF STOCK!")
    except Exception as e:
        print("Error:", e)

# ----- SMART SERVICE -----
def smart_service():
    vehicle = input("Vehicle Number: ")
    cursor.execute("SELECT purchase_date, current_km FROM customers WHERE
vehicle_number=%s", (vehicle,))
    result = cursor.fetchone()
    if not result:
        print("Vehicle not found!")
        return
    purchase_date, km = result
    today = datetime.today()
    months = (today.year - purchase_date.year) * 12 + today.month - purchase_date.month
    print("\nKM:", km, "| Months:", months)
    print("\nAvailable Services:")
    print("Basic, Oil Change, Full, Major")
    if km > 20000 or months > 24:
        print("Recommended: Major Service")
    elif km > 10000 or months > 12:
        print("Recommended: Full Service")
    elif km > 5000 or months > 6:
        print("Recommended: Oil Change")
    else:
        print("Recommended: Basic Service")

# ----- PDF BILL -----
def generate_pdf_bill():
    vehicle = input("Enter Vehicle Number: ")
    cursor.execute("SELECT customer_id, name FROM customers WHERE vehicle_number=%s",
(vehicle,))

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data = cursor.fetchone()
if not data:
    print("Customer not found!")
    return
cid, name = data
service_cost = float(input("Service Cost: "))
parts_details = []
total_parts_cost = 0
while True:
    use = input("Use spare part? (y/n): ")
    if use.lower() != 'y':
        break
    part_id = int(input("Part ID: "))
    qty_used = int(input("Quantity Used: "))
    cursor.execute("SELECT quantity, price, part_name FROM spare_parts WHERE
part_id=%s", (part_id,))
    part = cursor.fetchone()
    if not part:
        print("Part not found!")
        continue
    stock, price, part_name = part
    if qty_used > stock:
        print("Not enough stock!")
        continue
    new_stock = stock - qty_used
    cursor.execute("UPDATE spare_parts SET quantity=%s WHERE part_id=%s",
(new_stock, part_id))
    cost = price * qty_used
    total_parts_cost += cost
    parts_details.append((part_name, qty_used, price, cost))
    print(f"{part_name} used | Cost: ₹{cost}")
    if new_stock < 5:
        print(f"LOW STOCK: {part_name} only {new_stock} left!")
    if new_stock == 0:
        print(f"{part_name} OUT OF STOCK!")

total = service_cost + total_parts_cost
gst = total * 0.18
final = total + gst
cursor.execute("INSERT INTO bills VALUES (NULL,%s,%s,%s,%s,NOW())",
                (cid, total, gst, final))
conn.commit()

# ----- PDF GENERATION -----
doc = SimpleDocTemplate(f"bill_{cid}.pdf")
styles = getSampleStyleSheet()
content = []

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content.append(Paragraph("Vehicle Service Bill", styles['Title']))
content.append(Spacer(1, 15))
content.append(Paragraph(f"Customer: {name}", styles['Normal']))
content.append(Paragraph(f"Vehicle: {vehicle}", styles['Normal']))
content.append(Spacer(1, 10))
content.append(Paragraph("---- Service Charges ----", styles['Heading3']))
content.append(Paragraph(f"Service Cost: ₹{service_cost}", styles['Normal']))
content.append(Spacer(1, 10))
content.append(Paragraph("---- Spare Parts Used ----", styles['Heading3']))
if not parts_details:
    content.append(Paragraph("No spare parts used", styles['Normal']))
else:
    for part in parts_details:
        pname, qty, price, cost = part
        content.append(Paragraph(
            f"{pname} | Qty: {qty} | Price: ₹{price} | Cost: ₹{cost}",
            styles['Normal']
        ))
    content.append(Spacer(1, 10))
    content.append(Paragraph(f"Total Parts Cost: ₹{total_parts_cost}",
styles['Normal']))
    content.append(Spacer(1, 10))
    content.append(Paragraph("---- Billing ----", styles['Heading3']))
    content.append(Paragraph(f"Subtotal: ₹{total}", styles['Normal']))
    content.append(Paragraph(f"GST (18%): ₹{gst}", styles['Normal']))
    content.append(Paragraph(f"Final Amount: ₹{final}", styles['Normal']))
doc.build(content)
os.startfile(f"bill_{cid}.pdf")
print("PDF Generated with full details!")

# ----- MENU -----
def main():
    while True:
        print("\n===== VEHICLE SERVICE SYSTEM =====")
        print("1 Add Customer")
        print("2 View Customers")
        print("3 Search Customer")
        print("4 Update KM")
        print("5 Book Service")
        print("6 Update Service")
        print("7 Service History")
        print("8 Add Part")
        print("9 View Parts")
        print("10 Update Part Stock")
        print("11 Smart Service")
        print("12 PDF Bill")
        print("0 Exit")

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ch = input("Choice: ")
if ch == "1": add_customer()
elif ch == "2": view_customers()
elif ch == "3": search_customer()
elif ch == "4": update_km()
elif ch == "5": book_service()
elif ch == "6": update_service()
elif ch == "7": service_history()
elif ch == "8": add_part()
elif ch == "9": view_parts()
elif ch == "10": update_part_stock()
elif ch == "11": smart_service()
elif ch == "12": generate_pdf_bill()
elif ch == "0": break
else: print("Invalid")

if __name__ == "__main__":
    main()
```